Smart Metro Ticket Booking System

Requirement Analysis Phase

Team ID

Project Name: Smart Metro Booking System

Maximum Marks: 4 Marks

3. Solution Requirements

This document brings together the functional and non-functional requirements for the ServiceNow-based Metro Ticket Booking solution, along with component mapping, actor roles, assumptions, and the delivery milestones agreed for this phase.

3.1 Functional Requirements

Ticket Booking (Service Catalog)

- Provide a Service Catalog item for booking with variables: Source Station, Destination Station, Passenger Type, and Number of Tickets.
- Validate the station selection against Station Master to block invalid routes.
- Support General, Student, and Senior Citizen passenger types.
- Allow multiple tickets to be booked in a single transaction.

Fare Calculation

- Calculate fare automatically based on the distance/zone between source and destination.
- Apply passenger-type discounts and multi-ticket totals automatically.
- Show an itemized fare breakup before payment is confirmed.

Payment Processing

- Integrate UPI, debit/credit card, and mobile wallet payment modes.
- Confirm payment status back to the booking flow in real time.
- Log every transaction with a unique reference number for audit and reconciliation.

QR Ticket Generation

- Generate a unique QR-coded digital ticket right after successful payment.
- Make each QR code single-use and time-bound to the booked journey.
- Store the generated ticket against the passenger's booking record for later retrieval.

Notifications

- Deliver the digital ticket and QR code via in-app notification, email, and WhatsApp/SMS.
- Notify passengers of payment failures with a retry link.
- Alert Metro Operations of payment exceptions or QR generation failures.

Administration & Reporting

- Let Station Managers maintain station and fare master data through a controlled workflow.
- Give Metro Operations dashboards on ticket volume, revenue, and peak usage.
- Let IT Admins configure catalog variables, ACLs, and integration endpoints.

Figure 3.1 — End-to-end flow across the functional requirement areas above

3.2 Non-Functional Requirements

Category	Requirement
Performance	Fare calculation and QR generation must complete within 3 seconds of payment confirmation under normal load.

Category	Requirement
Scalability	The solution must handle concurrent booking spikes during peak commute hours without degrading.
Availability	Core booking and QR validation services should target 99.5% uptime during metro operating hours.
Security	All payment data must go through PCI-compliant gateways; role-based access control (ACLs) must govern every sensitive table.
Usability	A first-time user should be able to complete booking in under 5 steps, with accessible, mobile-friendly UI Pages.
Auditability	Every booking, payment, and QR generation event must be logged with a timestamp and correlation ID for traceability.
Maintainability	Fare rules and station data must be configurable by IT Admins without a code deployment.

3.3 ServiceNow Component Mapping

Requirement Area	ServiceNow Component	Purpose
Ticket Booking Form	Service Catalog + Catalog Client Scripts	Captures Source, Destination, Passenger Type, and Number of Tickets with real-time validation.
Fare Calculation	Business Rules / Script Includes	Runs fare logic server-side based on station distance and passenger type.
Booking-to-Ticket Workflow	Flow Designer	Orchestrates the end-to-end flow: payment trigger, QR generation, notification dispatch.
Payment Integration	Integration Hub / REST Spokes	Connects to external payment gateway APIs for UPI, card, and wallet transactions.
QR Code Generation	Integration Hub / Script Include (QR library)	Generates a unique QR image and links it to the ticket record.
Alerts & Ticket Delivery	Notifications (Email/SMS/Push)	Delivers digital tickets and status alerts to passengers and staff.
Passenger Self-Service	Service Portal / UI Pages	Hosts the booking form, fare preview, booking history, and QR ticket display.
Reporting	Performance Analytics / Reports	Gives Metro Operations usage, revenue, and peak-time dashboards.

Figure 3.2 — Layered ServiceNow architecture realizing the component mapping above

3.4 Actors & Roles

Actor	Role in the System
Passengers (End Users)	Book tickets, make payments, receive and use QR tickets, view booking history.
Station Managers	Maintain station/fare master data, validate QR tickets at gates, monitor station-level ticket volume.
Metro Operations Team	Consume analytics and reports, manage exceptions, plan schedules based on demand data.
IT Admins	Configure catalog items, flows, integrations, ACLs, and troubleshoot platform issues.

3.5 Delivery Milestones

#	Milestone	Description
1	Catalog Form Setup	Build the Service Catalog item with Source, Destination, Passenger Type, and Number of Tickets variables.

#	Milestone	Description
2	Fare Logic Script	Implement server-side fare calculation covering distance/zone and passenger-type rules.
3	QR Code Integration	Integrate a QR generation library/API and link its output to the ticket record.
4	Station Details Mapping	Load and map the Station Master data used for route validation and fare lookup.
5	UI Pages	Design and build passenger-facing Service Portal pages for booking, fare preview, and ticket display.
6	UAT & Deployment	Run user acceptance testing with all personas and deploy to production.

3.6 Assumptions & Constraints

- Passengers have access to a smartphone or a WhatsApp-enabled device for receiving digital tickets.
- A payment gateway supporting UPI, card, and wallet is available for integration.
- The Metro Operations Team owns and maintains station and fare master data.
- QR-based Automated Fare Collection (AFC) gate hardware is available or planned at stations for scanning.